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Robo-advisors in financial services: Redefining wealth management in the age of artificial intelligence

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ABSTRACT

In recent years, the delivery and accessibility of investment advisory services have been redefined through the integration of Financial Services and Artificial Intelligence (AI). The present study develops a structured conceptual model, grounded in Dubin's (1978) framework on theory-building, to understand how distinct institutional designs of robo-advisors: Wealth Management Firm Robo-Advisors (WMRAs), Regulatory Robo-Advisors (RRAs), and Personal Robo-Advisors (PRAs), contribute to three core outcomes: return maximization, risk minimization, and financial inclusion. These outcomes form the heart of the conceptual model, highlighting how institutional form determines strategic value.

The present study emphasizes the role of behavioral and contextual mechanisms, including trust, digital competence, compliance orientation, and perceived security, as well as moderators such as platform type and institutional constraints, rather than focusing solely on technological effectiveness. By integrating these mechanisms with institutional design and market conditions, the framework examines the interplay between robo-advisor structures and their outcomes.

The paper introduces a temporal perspective, i.e., how regulation and user competence change over time through feedback loops, and clearly specifies situations in which the proposed relationships may fail or be empirically disproven. The study serves as a foundation for empirical testing and experimental studies, with significant implications for policy design, innovation, and regulatory governance in financial advisory services.

1. Introduction

Financial services are a cornerstone of economic systems, encompassing banking, insurance, investment, asset management, and advisory solutions (Gomber *et al.*, 2018). Human financial advisors have traditionally played a vital role in guiding investment decisions, offering tailored financial planning, managing risk preferences, and providing emotional reassurance during volatile market conditions. However, the human-centric model is not without limitations, as human advisors are constrained by cognitive biases, limited scalability, and inconsistent service quality across clients (Jung et al., 2019; Kahneman, 2011). The availability of traditional advisory services is often bound by geography and time, and personalized advice can be unaffordable, particularly for smaller investors or under-resourced institutions (Jung *et al.*, 2018).

In response to these limitations, the financial sector has undergone a transformation through the integration of digital technologies into its services. AI has become a pivotal force, enhancing efficiency, fostering automation and personalization, thereby reshaping decision-making processes and customer engagement across sectors, including marketing (Davenport et al., 2020; Verma et al., 2021) and healthcare (De Fauw et al., 2018), among others.

The use of AI in financial services facilitates tasks such as credit scoring, fraud detection, and automated portfolio management. A prominent application of the technological shift is the emergence of Robo-Advisors, which provide personalized financial advice for investment management (Jung et al., 2018). Robo-advisors utilize data analytics, behavioral insights, and market signals to provide low-cost, scalable, and often unbiased alternatives to traditional financial advisors.

In the B2C domain, robo-advisors have been extensively adopted to

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establish a logically consistent conceptual foundation. Fig. 1 below helps in better understanding the phases and logical flow of Dubin's Theory-Building framework.

The conceptual framework rests on three major assumptions. Firstly, the study assumes a baseline level of digital competence among the users. Secondly, it assumes a minimum threshold of institutional and regulatory readiness (e.g., data protection norms, regulatory standards) to enable the governance functions implied in the present study. Lastly, it is expected that users and organizations act under bounded rationality, where perceptions of cost–benefit, trust, and biases are central to adoption decisions. These assumptions clarify the scope and prevent the framework from being overextended into unsuitable contexts.

The present study conducted a literature review using the keywords - *"Robo" and "Advisor"*, and got a total of *2592* results, which was further filtered down to the work done only in the area of – "Business, Management and Accounting", "Finance, Economics and Econometrics" as well as "Decision Sciences" as the previous work done in these domains would be most helpful in further building the theory. The filter resulted in a total of 399 Search Results, out of which 154 papers were further shortlisted (including both articles and review papers) based on the study's scope, the journal's quality, and Relevance to the Research Topic. The study then filtered out any proceedings or conference/ workshop presentations from the *399* results. Next, a thematic analysis was conducted on the 154 shortlisted papers using the ADO (Antecedents, Drivers, and Outcomes) framework, resulting in the selection of 27 shortlisted papers.

2.1. Units of the theory

The study synthesizes the extant literature on "Robo-Advisors" using scientific databases like Scopus. Systematic literature protocols, using the PRISMA framework, facilitated the retrieval of relevant literature. The thematic analysis of retrieved literature, using the ADO framework, discerned constructs, interaction mechanisms, and outcome patterns relevant to robo-advisory systems and AI-enabled financial services (Nain and Rajan, 2024). For institutional robo-advisors, decision-making authority is distributed across multiple organizational actors rather than focusing on an individual investor. Strategic investment objectives are typically defined at the top management level, while being executed by portfolio managers, which leads to a gap between regulatory and compliance requirements (Capponi et al., 2020; Jung *et al.*, 2018). Coleman (1990) presented a chain of reasoning to justify aggregating individual choices into organizational choices. Individual rational choice, informed by a common sociological conception, translates into collective action, and thus, individual behavior becomes organizational behavior (Coleman, 1990). According to Foss and Mazzelli (2025), the organizational objectives of return maximization and risk minimization are not exogenous functions, but rather they emanate from the collective aspirations of individuals within the organization.

Organisations should not be treated as a single entity, but rather as a collective aggregation of their units. Organisations should be studied as interfaces between their actors and social mechanisms (Foss & Mazzelli, 2025). Collective organizational goals (risks and income) emerge through a dominant coalition (Cyert & March, 1963). A dominant coalition is an informal group that is not characterized by hierarchical positions (e.g., CEO, employees, or regulators); rather, shared goals and motivations drive the organization's expected outcomes (Mithani & O'Brien, 2021). At times, active upward communication from employees to top management (CEO) shapes the organization's priorities (Kor & Mesko, 2013; Ma et al., 2020). The organization's risk profile is not static; it is formed through the dynamic interplay of individual traits and social mechanisms operating through communication (Foss & Klein, 2022; Mithani & O'Brien, 2021).

B2C is distinct from B2B, where investment decisions are individual-based, explicit, and directly observable (Belanche *et al.*, 2025). Based on the investment context, i.e., B2B (Institutional Investment) or B2C (Individual Investment) or Regulatory context, robo-advisors are primarily classified into three types serving the focused needs of each context as depicted in Table 1. Additionally, the existing literature identifies three primary desired outcomes expected across various contexts: return maximization, risk minimization, and financial inclusion, which are further detailed in Table 1. The outcomes are measurable and performance oriented. The outcomes map the systemic impact of individual-unit level changes in the adoption of robo-advisors and the subsequent impact on the financial ecosystem (Fatima and Chakraborty, 2024; Matthews *et al.*, 2025; Nain and Rajan, 2024).

2.1.1. Types of Robo-advisors

The three forms of robo-advisors, classified contextually, have distinct characteristics and objectives. Wealth management robo-advisors (WMRAs) primarily operate within institutional investment environments, such as wealth management firms and asset management companies. The functions of WMRAs are embedded within organizational structures rather than individual investor preferences, integrating institutional inputs, such as portfolio strategies defined by fund managers, governance protocols, and regulatory constraints within firms. As a result, the decision-making is standardized and formalized through algorithmic rules that ensure consistency across portfolios and clients. WMRA platforms support scalability and coordination within complex institutional settings by translating heterogeneous operational inputs into uniform advisory processes that align with firm-level policies and compliance requirements (Boreiko et al., 2020; Capponi et al., 2020).

Regulatory Robo-Advisors (RRA) focus on supervisory and compliance factors and are primarily oriented toward embedding regulatory logic into automated financial advisory services. These systems support regulatory functions by enabling continuous monitoring, standardization, and auditability of algorithmic advisory practices. RRA frameworks shape how robo-advisory platforms are governed and deployed by formalizing requirements related to transparency, accountability, and data protection. In doing so, RRAs establish the institutional boundaries within which automated advisories operate, and reinforce regulatory oversight in increasingly algorithm-driven financial environments (Reher and Sokolinski, 2024)

Personal Robo-Advisors (PRA) deliver automated financial advisory services directly to individual investors in a B2C context. These systems rely on digital interfaces, automated onboarding processes, and algorithmic profiling to assess user-specific characteristics such as financial goals, investment horizon, and investment tolerance parameters. PRAs focus on ease of access, personalization, and continuous engagement through low-touch digital channels. The advisory logic is embedded within algorithms that operate consistently across users, while individual differentiation is achieved through parameterized inputs rather than interpersonal discretion (Reher and Sokolinski, 2024; Zhu et al., 2025).

2.1.2. Expected Outcome

Prior studies have predominantly focused on adoption, engagement, and trust-related outcomes, emphasizing user acceptance and behavioral factors. Bashir *et al.* (2025) propounded that trust propensity, perceived system ability, and facilitating conditions influence the engagement with robo-advisors. In contrast, core financial outcomes such as return maximization, risk minimization, and financial inclusion receive comparatively limited attention despite being central objectives of financial advisory services. This imbalance underscores the need for a more outcome-oriented and system-level theorization of robo-advisors. The present study aims to bridge the gap by focusing on financial outcomes like return maximization, risk minimization, and financial inclusion.

Previous studies examined how algorithmic platforms, through data-intensive decisions, can outperform existing human advisors (Boreiko *et al.*, 2020; Capponi *et al.*, 2022). In the case of human advisory services, return maximization involves personalized investment strategies designed by financial advisors, who utilize historical trends, previous

Table 1
Types of robo-advisors and the impact on expected outcomes.

Types of Robo-Advisors

Expected Outcomes	Wealth Management Firm Robo-Advisors (WMRA) Potential Conflict of Interest	Regulatory Robo-Advisors (RRA) Regulatory alignment	Personal Robo-Advisors (PRA) Have platform autonomy
Maximize Returns	Utilize personalized models and AI to allocate assets and tailor solutions to meet specific customer goals and risk expectations.	Establish performance benchmarks for all products and implement standards across platforms to foster fair competition.	Offer micro-investment options and goal-based AI strategies for customised investing. Also, provide tax-efficient options.
Minimize Risk	Automate investment portfolios rebalancing and risk profiling using algorithms to reduce bias and emotional investing.	Conduct risk audits for client protection and regulate algorithmic transparency. Also, enforce investor protection laws across products.	Utilize machine-learning tools and real-time analytics to predict volatility and mitigate risk.
Financial Inclusion	Provide low-cost, advisory, and financial services to underserved and low-income populations. Offer simple, achievable yet profitable investment solutions to low-ticket clients.	Promote digital and financial literacy through targeted campaigns and incentives for key users. Provide incentives for inclusion, such as regulatory sandboxes and relaxation of e-KYC requirements.	Create mobile-first platforms with gamified onboarding to target young users. Additionally, multilingual Apps with 24/7 customer support and zero-balance products should be developed to encourage underserved segments.

experience, as well as professional judgment. Automated advisors, on the other hand, utilize algorithmic methods, such as asset rebalancing and predictive analytics, to enhance investment returns by relying on data-driven decision-making and optimizing risk-return trade-offs. The automated advisory platforms reduce bias and human errors, and robo-advisors can prevent reactionary investing through systematic approaches (Belanche *et al.*, 2025; Kahneman and Tversky, 1979). The elimination of behavioral biases through an automated advisory platform alters the outcomes (Kahneman, 2011). The expected return maximisation in institutional investment varies sharply from individual investment (Ardichvili *et al.*, 2003; Jung *et al.*, 2018), and WMFRA translates heterogeneous client mandates, managers' strategies, and regulatory constraints into standardized, algorithmically optimized investment decisions (Boreiko *et al.*, 2020; Capponi *et al.*, 2020). Unlike traditionally managed portfolios, robo-advisors dynamically adjust asset weights and capitalize on market inefficiencies to increase returns for users and achieve the users' investment goals (Capponi *et al.*, 2020).

Robo-advisors assess clients' risk appetite through historical data and utilize expertise to recommend strategies that align with investors' long-term goals. Robo-advisors also overcome the inconsistencies in decision-making and emotional reactions, as well as the vulnerabilities of human advisors (Kahneman and Tversky, 1979). Robo-Advisors reduce risks by enforcing disciplined diversification, applying risk thresholds, and reducing emotionally driven decisions (Belanche *et al.*, 2025). At the organizational level, regulatory compliance, internal risk limits, and portfolio diversification rules (Jung *et al.*, 2018; Reher and Sokolinski, 2024) dictate the investment decision-making risk profiles as quantitative indicators as formal investment mandates (Capponi *et al.*, 2020). Robo-Advisors operationalize these indicators by embedding them into algorithmic decision rules, ensuring consistency across portfolios while also maintaining institutional risk appetites (Boreiko *et al.*, 2020; Capponi *et al.*, 2020). Furthermore, the algorithmic approach helps protect investors from impulsive actions, such as panic selling, and promotes long-term consistency (Ahmad *et al.*, 2025). The structured approach of robo-advisors mitigates the effect of behavioral biases prevalent in traditional advisory services and aids in reducing panic-based decisions, particularly in volatile environments (Jung *et al.*, 2019; Zhu *et al.*, 2025).

According to traditional advisory systems, financial inclusion is constrained by high minimum investment requirements, as well as limited exposure and time constraints, which may lead to the exclusion of smaller investors. Hence, financial inclusion under human advisors has often remained narrow and geographically restrained. In comparison, robo-advisors have lower or no minimum investment requirements and reduced advisory costs, which makes them appealing to small-scale investors, underbanked segments, and especially to young investors who may not have access to the scale of funds required by traditional advisory services (Ha *et al.*, 2025). Recent empirical work by Bashir *et al.* (2025) highlighted the significance of robo-advisors in promoting financial inclusion in emerging economies. The accessibility of technology across social strata has facilitated the financial inclusion of underserved segments, including first-time investors and those with limited investable assets (Ha *et al.*, 2025). RAs also improve returns, especially for underserved groups, by improving responsiveness to market conditions and reducing friction (D'Hondt *et al.*, 2020; Zhu *et al.*, 2025). Hence, robo-advisors are being positioned as tools for enhancing financial inclusion, particularly in developing markets and within smaller organisations that previously had limited access to advisory platforms. By integrating algorithm-based personalization, automated onboarding, and behavioral factors, automated advisors promote

inclusive participation within the financial markets, offering lower fees, a lower minimum threshold, and digitally scalable platforms (Ha et al., 2025; Reher & Sokolinski, 2024).

2.1.3. Factors Leading to the Expected Outcomes

Both external factors (such as platform design and regulatory implications) and internal forces (like trust and digital literacy) contribute to the influence of robo-advisors in achieving the desired outcomes (Matthews *et al.*, 2025; Nain & Rajan, 2023; Seiler & Fanenbruck, 2021).

The extant literature explored the role of perceived ease of use and user experience in the adoption of robo-advisors (Jung *et al.*, 2018). Robo-advisors rely heavily on automation capabilities and AI-enabled financial modeling to reduce risk exposure by utilizing stable asset allocation methods (Fatima & Chakraborty, 2024). It also helps to enhance investment returns through real-time portfolio rebalancing (Matthews *et al.*, 2025). For underserved and financially excluded segments, easy and accessible onboarding, along with intuitive designs, play a crucial role in enhancing participation (Jung *et al.*, 2018). The platform-level designs help improve both system-level outcomes and individual user experiences when using robo-advisory platforms.

Market Conditions are another critical driver that influences investors' willingness to adopt robo-advisors (Jung et al., 2018). Market friction and financial complexity act as a trigger for algorithm-based advisory services (Matthews *et al.*, 2025). Volatility in the market, along with economic uncertainty and interest rate fluctuations, impacts the expected outcomes from both a financial and macroeconomic perspective. Capponi *et al.*, (2020) suggest that users are more likely to seek automated advice during uncertain or turbulent periods when unsure about the expected returns. Robo-advisors are better suited to support both return maximization and capital preservation by dynamically responding to shifts in market conditions through the application of risk filters and adjustments to asset allocations (Ahmad *et al.*, 2025).

Traditionally, human financial advisors have helped clients in investment decisions, manage wealth, and design portfolios tailored to specific client needs (Reher & Sokolinski, 2024), and have been integral in the wealth management system, providing personalized asset allocation and portfolio construction (Jung *et al.*, 2018; Kahneman, 2011). The limited cognitive abilities of humans and subjective biases limit the capabilities of human financial advisors, necessitating the need for robo-advisors. Robo-advisors are algorithm-driven, digital platforms that utilize analytics and machine learning to allocate assets, assess risk, and rebalance portfolios, while delivering both basic and complex financial services at a faster pace and with greater efficiency (Capponi *et al.*, 2020). Robo-advisors are favoured over human advisors for several reasons, including cost efficiencies (Belanche *et al.*, 2025), enhanced transparency and consistency, and accessibility (Ha *et al.*, 2025).

2.2. Laws of Interaction

The relationship between Units of the Theory is described by Laws of Interaction (Dubin, 1978). These laws outline how a change in one or more units of theory affects the remaining units or constructs (Ardichvili *et al.*, 2003). In the context of Robo-Advisors, the laws explain the mechanism through which the automated advisors lead to the final outcomes, i.e., Return Maximization, Risk Minimization, and Financial Inclusion. Previous studies in the field of automated advisory systems have focused on multiple factors affecting both the adoption and performance of robo-advisors in various contexts; however, these factors are often examined in isolation.

The functional capabilities of robo-advisors have been the primary drivers of strategic outcomes in many prior studies. Some of these technological factors include Automated Asset and Portfolio Rebalancing, Tax-Loss Harvesting, Compliance Monitoring, Real-Time Assessment, and Portfolio Optimization, which help them outperform traditional advisory systems. Such capabilities lead to increased scalability, reduced portfolio risk, and cost efficiency (Boreiko *et al.*, 2020; Capponi *et al.*, 2020). However, these analyses assume homogeneous digital readiness across users with less friction.

Robo-Advisors are not merely a function of technology, but an automated system that is mediated by behavioral and psychological constructs, such as cognitive biases, trust, and perceived value. Cognitive biases, such as anchoring, overconfidence, and status quo bias, affect investor judgment and may reduce engagement (Ahmad *et al.*, 2025). The users' trust in the robo-advisor's automated algorithm-based advice influences the adoption and outcome of the institutional financial service offering (Belanche *et al.*, 2025; Nourallah *et al.*, 2023). Bashir et al. (2025) emphasize the role of trust as a central mediating mechanism in the adoption and effectiveness of robo-advisors among young retail investors. The study demonstrates how trust operates as an intervening variable between platform attributes and behavioral outcomes, such as continued use and investor participation. In emerging markets, where digital adoption often outpaces financial literacy, trust serves as a stabilizing mechanism, enabling users to rely on algorithmic advice despite having limited prior experience. These insights complement the present study's laws of interaction by substantiating the behavioral pathways through which robo-advisory systems influence financial inclusion outcomes. Perceived value is influenced by factors such as personalization, convenience, and cost efficiency, versus trade-offs like data risks and a lack of human touch (Fatima & Chakraborty, 2024). These behavioral aspects represent a shift from the traditional techno-functional model to contemporary AI-enabled financial advisory business models.

Contextual variables, such as the type of robo-advisor, perceived security, and organisational features, moderate the phenomenon. Whether the advisory system is fully automated, bionic (hybrid), or simply regulatory aligned determines both user expectations and functionality (Jung *et al.*, 2018; Seiler & Fanenbruck, 2021). Factors such as data integrity and privacy concerns pivot the organisational decision to the degree of inclusion of automated advisory systems in financial services (Buhmann *et al.*, 2020; Nourallah *et al.*, 2023). Organizational features, such as investment strategy alignment, governance frameworks, and digital maturity, influence both the acceptance and effectiveness of automated advisors (Fatima & Chakraborty, 2024; Matthews *et al.*, 2025).

Understanding the interaction of mediating variables with one another is necessary to comprehend the feedback loops. For example, a higher level of trust can mitigate the adverse effects of scepticism among users in the context of automation. It also helps increase platform engagement, even in uncertain conditions (Nourallah *et al.*, 2023). Additionally, digital competence can increase perceived value by enabling better utilization of advanced features and reducing ambiguity (Belanche *et al.*, 2025). On the other hand, extreme behavioral biases may diminish the perceived value and reduce trust, potentially leading to the superficial or selective use of advisors (Ahmad *et al.*, 2025).

While existing studies provide limited insights into the functions of robo-advisors by examining the direct effect, behavioral mediators, and contextual moderators, most of them are rooted in the B2C context and primarily focus on a technology-first lens. Institutional variables, user scepticism, demographics, and typological distinction of robo-advisors are often treated as secondary variables. The present study, on the other hand, examines it from both B2C and B2B perspectives to discern both institutional and individual variables in adoption and outcomes of robo-advisors. Table 2 will further help in understanding the extension of the contribution from the present study.

The present study proposes a conceptual model (Fig. 2) that incorporates contextual considerations (behavioral and institutional factors) of Wealth Management Firm Robo-Advisors (WMRAs), Regulatory Robo-Advisors (RRAs), and Personal Robo-Advisors (PRAs) in achieving the desired outcomes (Return Maximization, Risk Minimization, and Financial Inclusion). The following laws of interaction capture the restructured logic based on the conceptual framework:

Table 2
Contributions of the study.

Interaction	Extant Literature	Present Study (Unique Additions)
Direct Effects	Functional efficiency of robo-advisors (general)	WMRA, RRA, and PRA types mapped to specific outcomes (maximize returns, minimize risk, and financial inclusion)
Mediators	Trust, perceived value, digital literacy	Explicit inclusion of institutional trust, cognitive biases, and benefit-cost evaluation in B2B logic
Moderators	Weakly discussed, mostly demographic	Detailed modelling of platform types, compliance orientation, and automation scepticism
Mediator Interactions	Often overlooked	Dynamic interplay between trust, scepticism, competence, and value perception

2.2.1. Law 1: Mediating variables translate technical capabilities into strategic outcomes

The effectiveness of robo-advisors is not solely determined by technological capacity. However, it is also influenced through a set of mediators that shape how users interpret and act upon the recommendations. Trust of users in both the platform and the institution providing those services results in sustained engagement as well as increased adoption. The perceived cost-benefit trade-off, including assessments of ease, efficiency, personalization, control, and affordability, drives adoption decisions. Digital competence enables users to decode, navigate, and apply algorithmic insights effectively, improving the likelihood of realizing expected outcomes. Behavioral biases (e.g., status quo bias, overconfidence, loss aversion) can distort risk perception, hinder adoption, or prevent rational use of the platform. These mediators operate across user types but manifest differently based on digital maturity, financial experience, and platform exposure.

2.2.2. Law 2: Moderators shape the stability and strength of mediated effects

The relationship between robo-advisor capabilities and user outcomes is further conditioned by contextual moderators. The type of robo-advisor acts as a structuring agent, setting platform expectations and driving outcomes. The study aims to determine how different types of robo-advisors exhibit varying outcome orientations. Broadly, robo-advisors are classified into three types based on the functional orientations, viz. Wealth Management Firm Robo-Advisors (WMRAs), Regulatory Robo-Advisors (RRAs), and Personal Robo-Advisors (PRAs). WMRAs focus on return maximization through personalization, utilizing goal-based investment strategies, predictive analytics, and algorithm-based optimization. RRAs focus on complying with regulations and mitigating risk by following transparency protocols, conducting automated audits, and implementing conservative portfolio rebalancing. PRAs focus on providing inclusive access through low-cost, low-threshold offerings, mobile-first onboarding, and multilingual support, among other initiatives.

The other moderating factors include perceived security, scepticism, institutional, and demographic factors. Institutional and demographic factors, such as regulatory environment, organizational governance, income level, or geographic access, influence trust formation and usability. Perceived security, particularly regarding data privacy and algorithmic transparency, either reinforces trust or inhibits platform reliance. Scepticism toward automation, driven by past experiences, perceived loss of control, or lack of explainability, can suppress otherwise positive evaluations of benefit and trust. These moderators operate systemically, influencing how the platform is experienced and evaluated across different users and settings.

2.2.3. Law 3: Mediator interactions are non-linear

The mediators within the model are not mutually exclusive; these operate interactively and produce either reinforcing or counteracting effects. Trust, perceived benefits, digital competence, and behavioral biases mediate the influence of robo-advisors on desired outcomes. Trust in Advisors and Financial Services counterbalances the skepticism among users, especially among people from older generations who were accustomed to using traditional advisory services. Trust also helps in sustaining engagement in the face of uncertainty. The perceived cost-benefit trade-off includes factors such as assessments of efficiency and ease, level of personalization, control, and affordability, all of which drive adoption decisions. Digital competence amplifies the impact of perceived benefits, resulting in more confident and comprehensive platform use across various categories of investors. Behavioral Biases, including cognitive and behavioral biases, may skew all rational considerations, limiting the adoption of automated platforms regardless of the cost advantages or technical design benefits. The interactivity among the mediators reinforces the need to view robo-advisor adoption not as a deterministic outcome, but as an adaptive engagement process, which is affected by contextual, cognitive, and emotional factors.

2.3. Boundaries of the theory

According to Dubin's (1978) theory-building approach, a theoretical framework must specify the limits within which its units, outcomes, and interactions are valid. In the case of financial applications, compliance operates at two distinct levels: (a) general digital/data regulations such as privacy, cybersecurity, and data protection, and (b) sector-specific financial regulations - KYC requirements, suitability checks, fiduciary obligations, and licensing norms. These collectively fall under institutional constraints in the framework. The boundary conditions used in this study are outlined below.

2.3.1. Regulatory Conditions

The applicability of suggested robo-advisor types is assumed only in systems where fundamental data protection laws and compliance standards are in place, and platform-level audit mechanisms are available to ensure the trust mechanisms and perceived value dynamics remain true. In the context of informal peer-to-peer apps, for example, the regulatory

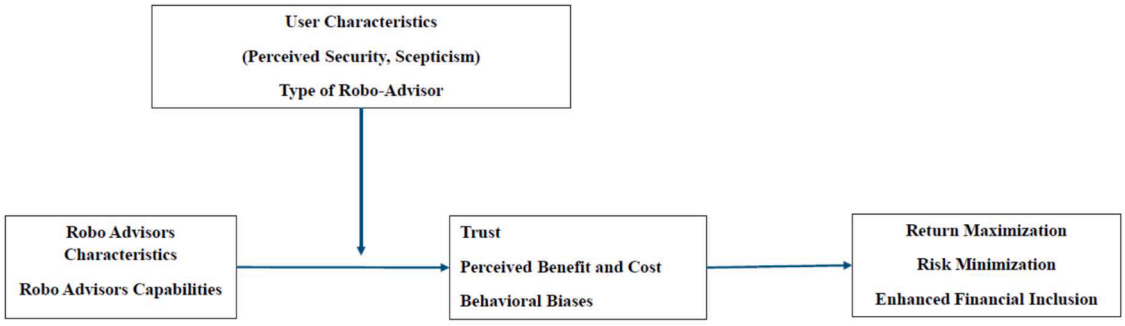

Fig. 2. Conceptual model.

conditions cannot be met, and hence the proposed theoretical framework may not be applicable.

2.3.2. Technological boundaries

According to the proposed model, the theory applies exclusively to algorithmic Robo-Advisory systems, which comprise three major types: Wealth Management Firm Robo-Advisors (WMRA), Regulatory Robo-Advisors (RRA), and Personal Robo-Advisors (PRA). Traditional advisors, chatbots without investment capabilities, and financial tracking tools fall outside the scope of the present study. The above boundaries align with previous studies that have focused on advanced functionalities, such as real-time decision-making, tax-loss harvesting, and portfolio optimization, among others (Capponi *et al.*, 2020).

2.3.3. User and institutional scope

The model includes aspects applicable to both B2B and B2C users of automated advisory services (robo-advisors). While the B2C context involves individual investors, the B2B context encompasses formal organizations, banks, or regulated intermediaries that rely on artificial intelligence, specifically AI-powered decision-making. However, it does not include informal or unregulated finance sectors, as the activities are not traceable through algorithmic pathways (Matthews *et al.*, 2025).

2.3.4. Relevance of outcomes

The theory holds only when robo-advisors are used for outcome-driven, formal, and intentional financial strategies, and not when the objective is merely to seek information or engage in passive monitoring. For the sake of simplicity and better understanding, the study considers three primary outcomes: Return Maximization, Risk Minimization, and Financial Inclusion as the expected outcomes (Fatima & Chakraborty, 2024; Belanche *et al.*, 2025).

2.4. System States

According to Dubin (1978), system states describe configurations of all units in the theory that persist over time and generate predictable patterns of outcomes. Extending the same to the present study and drawing insights from the works of Ardichvili et al. (2003) on entrepreneurial opportunity and Danielson's (2004) work on socialization for renewal, the study examines the system states based on three key features: inclusiveness, persistence, and distinctiveness.

2.4.1. Inclusiveness

The system state ensures that the theory is operational across different users, types of technologies, and institutional contexts. Three broader types of inclusivity include institutional, digital, and behavioural inclusivity. Institutional inclusivity applies to both emerging fintech startups that meet regulatory requirements and formal/ traditional financial institutions, such as banks, NBFCs, etc., which aligns with the need for structural legitimacy, as mentioned in Ardichvili *et al.* (2003). Previous works emphasize that the quality of digital infrastructure significantly affects inclusion outcomes (Fatima & Chakraborty, 2024). Digital infrastructure inclusivity must provide accessible and affordable interfaces and enable entry for both amateur and expert investors (Ardichvili *et al.*, 2003). Behavioral inclusivity accommodates varying levels of user trust, digital literacy, behavioral biases and user personalization, as is done in the socialization loops studied by Danielson (2004). User literacy and algorithmic trust mediate the adoption of platforms in both urban and rural settings (Ahmad *et al.*, 2025).

2.4.2. Persistence

It refers to the ability of the system to remain functional and relevant over a period through adaptive learning, changes, and constructive feedback. Feedback loops and learnings continuously evolve based on continuous feedback from users and the external environment. The above aspect is reflected in Danielson's (2004) argument for iterative organizational renewal. Feedback mechanisms in robo-advisors also contribute to better outcomes over time, as trust evolves. Strategic continuity for desired outcomes must align with long-term user goals, while also avoiding algorithmic drifts and other biases. Regulatory Co-Evolution, with regulatory bodies overseeing persistent systems, acts as a supporting factor that ensures financial innovation or technological advancement does not conflict with data privacy, fiduciary systems, or algorithmic fairness. Buhmann *et al.* (2020) emphasize how evolving institutional trust, along with transparency, leads to sustaining system resilience.

2.4.3. Distinctiveness

Highlights how a particular theory can operate under specific and non-trivial conditions and, more importantly, what makes it different from the existing theories and models, or what it adds to the existing knowledge base. Typological differentiation focuses on creating unique differentiation criteria and classifying Robo-Advisory systems into three distinct types: WMRA, RRA, and PRA systems. It acknowledges that the pathways to influence the expected outcomes vary significantly. Algorithmic-Behavioral Mediation is distinctive, and strategic outcomes arise when the decision-making process is differentiated through various perceptual and behavioral mediators. Cognitive perceptions directly impact algorithm acceptance among users (Ardichvili *et al.*, 2003; Belanche *et al.*, 2025)

2.5. Research propositions

Based on Dubin's (1978) framework, the following propositions are based on the interplay between the antecedents, like market conditions and technology and digital infrastructure/platform design, Mediators such as perceived cost-benefit, trust, etc., and moderators such as the type of robo-advisor, perceived security, etc., as are outlined by the laws of interaction. The propositions also consider the boundaries of the theory, as well as the applicable system states, ensuring relevance across different contexts.

Proposition 1. Advanced platform design and AI-driven analytics have a direct impact on the ability of automated advisors to maximize returns, minimize risks, and achieve financial inclusion (the expected outcomes of the study). When combined with high digital competence and positive cost-to-benefit analysis, technological quality transforms algorithmic capabilities into measurable strategic outcomes (Boreiko *et al.*, 2020; Capponi *et al.*, 2020).

***P1:** Better technological and platform design positively influences the expected outcomes, i.e., Return Maximization, Risk Minimization, and Financial Inclusion.*

Proposition 2. Volatile or restrictive market conditions heighten the value proposition of algorithmic advice by the demand for data-driven, objective decision-making. Adoption is further amplified when trust in the platform and perceived security are high, particularly among underserved populations and risk-sensitive investors (Ahmad *et al.*, 2025; Capponi et al., 2020).

***P2:** Market conditions such as volatility and access constraints influence how Robo-Advisors contribute to either Return Maximization, Risk Minimization, or Financial Inclusion.*

Proposition 3. Trust in the advisory platform itself, as well as the institution offering it, has a direct impact on sustained engagement, which is essential for realizing and maximizing long-term returns. Trust mitigates scepticism about automation, strengthens perceived value, and enables users to rely on systematic risk filters, thereby facilitating risk minimization. Also, trust lowers both institutional and psychological barriers for the underserved segments or first-time investors, thereby enhancing financial inclusion (Seiler & Fanenbruck, 2021).

***P3:** Trust acts as the mediator between antecedents and desired outcomes*

(return maximization, risk minimization, and financial inclusion).

Proposition 4. The user's ability to benefit from the platform features is enhanced via digital competence. It also aids in making informed decisions and interpreting algorithmic outputs, which leads to better allocation and return maximization. It also helps moderate the influence of behavioral biases and prevent panic-driven responses, thereby contributing to the minimization of risk. More importantly, higher competence reduces digital exclusion and increases investor participation, thereby enabling financial inclusion (Belanche *et al.*, 2025).

P4: *Higher digital competence mediates the relationship between technological quality and amplify desired outcomes.*

Proposition 5. Behavioral biases, such as loss aversion, status quo, or overconfidence, can diminish the impact of even well-designed platforms and distort usage adoption patterns, and may undermine return maximization or negatively impact risk minimization outcomes. Lower bias levels enable more rational interaction with robo-advisors, increasing the likelihood of achieving outcomes. Moreover, curbing biases, such as the fear of new technologies, increases the willingness among first-time and low-resource investors, thereby strengthening financial inclusion (Ahmad *et al.*, 2025; Kahneman and Tversky, 1979).

P5: *Lower behavioral biases strengthen the relationship between perceived benefit–cost trade-offs and desired outcomes.*

Proposition 6. Demographic factors such as age, income level, and prior investment experience shape adoption patterns and outcome realization among users. Younger, digitally literate users may respond more positively to innovation, which may lead to them seeing greater returns due to better engagement with advanced features. While the older generation and conservative investors may require additional trust-building and simplified features to minimize risk. Similarly, low-income groups benefit from affordable, digital-first access, which improves financial inclusion (Reher and Sokolinski, 2024).

P6: *User demographic characteristics, such as age, income, and investment experience, moderate the influence of robo-advisors on desired outcomes (return maximization, risk minimization, and financial inclusion).*

Proposition 7. Regulatory mandates, governance structures, and organizational risk appetite influence the adoption and integration of robo-advisors into decision-making processes. In practice, compliance operates at two levels: general digital regulations (e.g., privacy, cybersecurity) and sector-specific financial regulations (e.g., KYC checks, fiduciary obligations, suitability requirements, licensing norms). Institutional responsibility ensures that even personalized robo-advisors remain aligned with these regulatory standards. Strong regulations may aid in risk minimization, whereas governance and fiduciary responsibility can facilitate broader financial inclusion. Strategic alignment and a greater risk appetite may lead to return maximization, especially in the case of institutional investors (Matthews *et al.*, 2025).

P7: *Institutional constraints (like data compliance and financial regulations) moderate the influence of robo-advisors on desired outcomes (return maximization, risk minimization, and financial inclusion).*

The outlined propositions are amenable to operationalization using empirical techniques such as partial least squares structural equation modeling (PLS-SEM), which allows for both reflective and formative measurement models testing. However, consistent with the objectives of Dubin's framework and prior conceptual studies, the current manuscript is confined to theory construction and specification of relationships among variables.

3. Future research directions

The paper primarily aims to contextualize the potential of innovations like robo-advisors in contributing towards broader financial objectives, notably risk minimization, return maximization, and greater financial inclusion. Future studies could examine how conflicts of interest in wealth management robo-advisors may affect engagement, perceived trust, and adoption decisions, particularly when compared to regulatory robo-advisors that focus much more on regulation but may not necessarily give the maximum gains.

Examining user demographics, including generational cohorts, income levels, and digital literacy, as moderators of adoption and performance, can be another avenue for research. Similarly, institutional constraints such as regulatory flexibility and governance structures warrant deeper exploration to understand how these constraints shape usage patterns and outcome realization for different types of robo-advisory platforms.

Another important insight that emerges is the non-uniform influence of each contextual factor, also known as a moderating variable. The same robo-advisor may perform differently depending on the type of user, generational cohort, or the nature of use, thereby highlighting the requirement for a clear and nuanced understanding of the adoption drivers, which go far beyond the technology.

Future studies could also empirically examine how digital competence at the organizational level influences the trajectories of robo-advisor usage. Additionally, the role of institutional factors such as regulatory flexibility, automation culture, or multi-stakeholder decision-making frameworks warrants further exploration, especially when evaluating the performance of different robo-advisor types.

One could also focus on transparency and ethics, as Robo-Advisors become more autonomous and personalized. There will be a greater need to examine how algorithmic transparency, explainability, and fairness of AI-driven decisions affect trust, adoption, and societal equity in financial decisions.

Future research may empirically validate the proposed framework using quantitative methodologies such as partial least squares structural equation modeling (PLS-SEM), which is well-suited for models involving multiple mediators, moderators, and both formative and reflective constructs. Survey-based or archival data collected from institutional investors, wealth management firms, regulators, or individual users could be employed to test the proposed propositions and assess robustness across B2B and B2C contexts.

There is also an enormous scope for interdisciplinary research, especially by integrating AI ethics and Behavioral aspects, and the impact on decision-making across areas. Another unexplored area to investigate is the intersection between sustainable investing and robo-advisory, which presents considerable scope for research.

4. Contributions of the paper

The paper contributes to the conceptual understanding of robo-advisors by framing the adoption and expected outcomes within a structured theoretical model grounded in Dubin's theory-building methodology. Rather than merely categorizing robo-advisors or reviewing adoption factors in isolation, the framework integrates multiple interacting elements—types of platforms, user behavior, demographic factors, contextual conditions, and outcome dimensions—into a conceptual model. The model provides an institutional design of three distinct types of robo-advisors—Wealth-Management Firm Robo-Advisor (WMRA), Regulatory Robo-Advisor (RRA), and Personal Robo-Advisor (PRA)—and systematically maps them to outcomes of return maximization, risk minimization, and financial inclusion.

Unlike existing studies that primarily focus on technological efficiency or innovation, the framework expands the analysis to include behavioral, demographic, and contextual factors that influence the use of robo-advisors. By integrating these factors with institutional structures and market conditions, the paper provides a more comprehensive understanding of how robo-advisors impact investment decisions. The contribution lies not in reasserting innovation narratives but in offering a conceptual model that can guide empirical inquiry.

The paper focuses on how institutional risk profiles in B2B robo-advisory settings are formed through regulatory constraints, standardized risk metrics, and governance structures, rather than individual subjective preferences (Capponi *et al.*, 2020; Jung *et al.*, 2018). By distinguishing organizational risk formation from individual risk tolerance, the framework extends the research on advisory services (particularly robo-advisors) beyond predominantly retail-focused contexts and further contributes to a more institutionally grounded understanding of AI-enabled advisory systems (Boreiko *et al.*, 2020; Reher and Sokolinski, 2024).

There are several implications for the B2B services ecosystem. Wealth management firms can leverage Robo-advisors to restructure the resource integration process, combining algorithmic insights with human expertise to deliver value-added propositions tailored for different customers. Secondly, robo-advisors can create subscription-based or performance-linked pricing models to facilitate recurring revenue streams and deepen client relationships.

From a governance and risk perspective, data governance and algorithmic transparency emerge as prerequisites for sustaining trust, especially in institutional contexts with strict compliance requirements. These factors also directly influence adoption in B2C contexts, where demographic variables such as age, income level, and digital literacy mediate the perception of trust and value. Future studies can further apply the model in various socio-economic and regulatory contexts to test how different types of robo-advisors perform under different institutional and behavioral settings.

5. Limitations of the Study

Although the study provides a useful, structured conceptual model of robo-advisory adoption and outcomes in financial services, it has several limitations. First, as a conceptual and theory-building paper, the propositions are amenable for empirical testing. Future researchers may use descriptive or experimental designs to test the propounded propositions. Although the framework draws on extensive literature (Belanche *et al.*, 2025; Capponi *et al.*, 2020), real-world validation is necessary to assess the proposed relationships. Second, the framework assumes a minimum level of digital competence and institutional readiness across both institutional (B2B) and individual (B2C) users, while the degree of these constructs may vary across geographies and contexts. Third, the study primarily focuses on wealth management and portfolio advisory services within the digital finance sector. Other financial services, such as lending, payments, or insurance, are not examined in the present study, and future researchers may explore these unexplored territories. Fourth, the conceptualization of behavioral biases and scepticism is relatively broad. Future research could further refine these constructs by distinguishing between different forms of biases (e.g., anchoring, overconfidence, status quo) and skepticism (e.g., technology-based vs. institutional), which may exert distinct effects on both B2B and B2C adoption. The model assumes a rational evaluation of benefits and costs in decision-making and excludes factors such as organizational politics, cultural factors, or external shocks (e.g., regulatory changes, market crises).

6. Conclusion

The study advances a structured theory of robo-advisory adoption and outcomes, using Dubin's (1978) framework for theory building. The institutional typology highlights how robo-advisors are not just uniform technological solutions but contextually embedded structures that deliver distinct value depending on design and governance orientation. For example, B2C is distinct from B2B, where investment decisions are individual-based, explicit, and directly observable (Belanche *et al.*, 2025). At the organizational level, regulatory compliance, internal risk limits, and portfolio diversification rules (Jung *et al.*, 2018; Reher and Sokolinski, 2024) dictate the investment decision-making mandates (Capponi et al., 2020). Robo-Advisors operationalize these indicators by embedding them into algorithmic decision rules, ensuring consistency across portfolios while also maintaining institutional risk appetites (Boreiko *et al.*, 2020; Capponi *et al.*, 2020). Based on the investment context, i.e., B2B (Institutional Investment) or B2C (Individual Investment), or the Regulatory context, robo-advisors are primarily classified into three types, serving the focused needs of each context. At its core, the model differentiates three institutional types of robo-advisors—WMRAs, RRAs, and PRAs —and systematically maps them to three key outcomes: return maximization, risk minimization, and financial inclusion.

The framework further integrates behavioral mediators (e.g., trust, perceived cost–benefit, digital competence) and contextual moderators (e.g., platform type, perceived security) to explain how outcomes are realized in practice. These interactions highlight that technology alone does not drive impact; rather, it is the interaction between institutional design, user behavior, and regulatory environment that enables robo-advisors to create value.

The contribution is threefold: firstly, the study provides a novel, institutionally grounded model of robo-advisors; secondly, it establishes empirically testable propositions that future research can validate; and lastly, it offers actionable insights for regulators, policymakers, and practitioners seeking to balance innovation, inclusion, and governance.

Through the institutional framing and embedding it with Dubin's structured approach to theory-building, the paper positions robo-advisory systems not only as technological tools but also as evolving institutional forms capable of shaping financial ecosystems, which further opens up pathways for future research, experimental testing, and policy design, ultimately leading to the development of inclusive, resilient, and trust-based automated advisory systems.

CRediT authorship contribution statement

Janvi P. Shetty: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Poonam Singh:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Sanjeev Verma:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

Ahmad, U., Van Keulen, M., Briassouli, A., & Saad, M. (2025). Cognitive biases, Robo advisor and investment decision psychology: An investor's perspective from the New York Stock Exchange. *Acta Psychologica, 256*, Article 105048. https://doi.org/10.1016/j.actpsy.2025.105048

Ardichvili, A., Cardozo, R., & Ray, S. (2003). A theory of entrepreneurial opportunity identification and development. *Journal of Business Venturing, 18*(1), 105–123. https://doi.org/10.1016/S0883-9026(01)00068-4

Bashir, Z., Ali, F. H., Munir, S., & Takreem, K. (2023). Navigating the new frontier by mapping the intersection of financial, investment, and robo advice for individual, retail, and household investors: A bibliometric analysis. *Journal of Business and Management Research, 2*(2), 299–330. https://jbmrjournal.com/index.php/Journal/article/view/38.

Bashir, Z., Farooq, S., Iqbal, M. S., & Aamir, M. (2025). The role of trust in financial robo-advisory adoption: A case of young retail investors in Pakistan. *Sustainable Futures, 9*, Article 100538. https://doi.org/10.1016/j.sftr.2025.100538

Belanche, D., Casaló, L. V., Flavián, M., & Loureiro, S. M. C. (2025). Benefit versus risk: A behavioral model for using robo-advisors. *The Service Industries Journal, 45*(1), 132–154. https://doi.org/10.1080/02642069.2023.2176485

Boreiko, D., & Massarotti, F. (2020). How risk profiles of investors affect robo-advised portfolios. *Frontiers in Artificial Intelligence, 3*. https://doi.org/10.3389/frai.2020.00060. Article 60.

Buhmann, A., Paßmann, J., & Fieseler, C. (2020). Managing algorithmic accountability: Balancing reputational concerns, engagement strategies, and the potential of rational discourse. *Journal of Business Ethics, 163*, 265–280. https://doi.org/10.1007/s10551-019-04226-4

Capponi, Agostino and Olafsson, Sveinn and Zariphopoulou, Thaleia, Personalized Robo-Advising: Enhancing Investment through Client Interactions (December 21, 2020). Forthcoming in Management Science. doi:10.2139/ssrn.3453975.

Coleman, J. S. (1990). Commentary: Social institutions and social theory. *American Sociological Review, 55*(3), 333–339.

Cyert, R. M., & March, J. G. (1963). *A behavioral theory of the firm*. Englewood Cliffs, NJ: Prentice-Hall.

D'Hondt, C., De Winne, R., Ghysels, E., & Raymond, S. (2020). Artificial intelligence alter egos: Who might benefit from robo-investing? *Journal of Empirical Finance, 59*, 278–291. https://doi.org/10.1016/j.jempfin.2020.10.002

Danielson, M. M. (2004). A theory of continuous socialization for organizational renewal. *Human Resource Development Review, 3*(4), 354–384. https://doi.org/10.1177/1534484304271528 (Original work published 2004).

Davenport, T., Guha, A., Grewal, D., et al. (2020). How artificial intelligence will change the future of marketing. *The Journal of the Academy of Marketing Science, 48*, 24–42. https://doi.org/10.1007/s11747-019-00696-0

De Fauw, J., Ledsam, J. R., Romera-Paredes, B., et al. (2018). Clinically applicable deep learning for diagnosis and referral in retinal disease. *Nature Medicine, 24*, 1342–1350. https://doi.org/10.1038/s41591-018-0107-6

Ha, D., Le, P., & Nguyen, D. K. (2025). Financial inclusion and FinTech: A state-of-the-art systematic literature review. *Financial Innovation, 11*, 69. https://doi.org/10.1186/s40854-024-00741-0

Dubin, R. (1978). *Theory building* (2nd ed.). New York: The Free Press.

Fatima, S., & Chakraborty, M. (2024). Adoption of AI in financial services: The case of robo-advisors in India. *IIMB Management Review, 36*(2), 113–121. https://doi.org/10.1016/j.iimb.2024.04.002

Foss, N. J., & Mazzelli, A. (2025). Bringing managers and management back into strategy: Interfaces and dynamic managerial capabilities. *Journal of Business Research, 186*, Article 114947. https://doi.org/10.1016/j.jbusres.2024.114947

Foss, N. J., & Klein, P. G. (2022). *Why managers matter: The perils of the bossless company*. New York: Hachette.

Gomber, P., Kauffman, R. J., Parker, C., & Weber, B. W. (2018). On the FinTech revolution: Interpreting the forces of innovation, disruption, and transformation in financial services. *Journal of Management Information Systems, 35*(1), 220–265. https://doi.org/10.1080/07421222.2018.1440766

Jung, D., Dorner, V., Glaser, F., & Morana, S. (2018). Robo-advisory: Digitalization and automation of financial advisory. *Business and Information Systems Engineering, 60*(1), 81–86. https://doi.org/10.1007/s12599-018-0521-9

Jung, D., Glaser, F., & Köpplin, W. (2019). Robo-advisory: Opportunities and risks for the future of financial advisory. In V. Nissen (Ed.), *Advances in consulting research. Contributions to management science*. Cham: Springer. https://doi.org/10.1007/978-3-319-95999-3_20.

Kahneman, D. (2011). *Thinking, fast and slow*. New York: Farrar, Straus and Giroux.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica, 47*(2), 263–291. https://doi.org/10.2307/1914185

Kor, Y. Y., & Mesko, A. (2013). Dynamic managerial capabilities: Configuration and orchestration of top executives' capabilities and the firm's dominant logic. *Strategic Management Journal, 34*, 233–244.

Kowalkowski, C., Wirtz, J., & Ehret, M. (2024). Digital service innovation in B2B markets. *Journal of Service Management, 35*(2), 280–305. https://doi.org/10.1108/JOSM-12-2022-0403

Ma, S., Kor, Y. Y., & Seidl, D. (2020). CEO advice seeking: An integrative framework and research agenda. *Journal of Management, 46*, 771–805.

Matthews, M. J., Su, R., Yonish, L., McClean, S., Koopman, J., & Yam, K. C. (2025). A review of artificial intelligence, algorithms, and robots through the lens of stakeholder theory. *Journal of Management, 51*(6), 2627–2676. https://doi.org/10.1177/01492063241311855 (Original work published 2025).

Mithani, M. A., & O'Brien, J. P (2021). So what exactly is a "coalition" within an organization? A review and organizing framework. *Journal of Management, 47*(1), 171–206.

Nain, I., & Rajan, S. (2023). Algorithms for better decision-making: A qualitative study exploring the landscape of robo-advisors in India. *Managerial Finance, 49*(11), 1750–1761. https://doi.org/10.1108/MF-01-2023-0055, 24 October 2023.

Nain, I., & Rajan, S. (2024). A scoping review on the factors affecting the adoption of robo-advisors for financial decision-making. *Scientific Papers of the University of Pardubice, Series D: Faculty of Economics and Administration, 32*(1), 1884. https://doi.org/10.46585/sp32011884

Nourallah, M., Öhman, P., & Amin, M. (2023). No trust, no use: Exploring trust-building elements in robo-advisory. *Journal of Financial Reporting and Accounting, 21*(1), 60–83. https://doi.org/10.1108/JFRA-12-2021-0451

Reher, M., & Sokolinski, S. (2024). Robo-advisors and access to wealth management. *Journal of Financial Economics, 155*, 1–20. https://doi.org/10.2139/ssrn.3515707

Seiler, V., & Fanenbruck, K. M. (2021). Acceptance of digital investment solutions: The case of robo advisory in Germany. *Research in International Business and Finance, 58*, Article 101490. https://doi.org/10.1016/j.ribaf.2021.101490

Verma, S., Sharma, R., Deb, S., & Maitra, D. (2021). Artificial intelligence in marketing: Systematic review and future research direction. *International Journal of Information Management Data Insights, 1*(1), Article 100002. https://doi.org/10.1016/j.jjimei.2020.100002

Zhu, Q., Han, C., Liu, S., Li, Y., & Che, J. (2025). Artificial intelligence-driven financial innovation: A robo-advisor system for robust returns across diversified markets. *Expert Systems with Applications, 274*, Article 119090. https://doi.org/10.1016/j.eswa.2025.126881